



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

90352 – Machine Learning for Economists

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Final Project

Measuring Financial Connectedness among Emerging Equity Markets:

Diebold–Yilmaz, NETS and FNETS Compared

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Summary of the Base Paper and of the Network Estimators

Diebold and Yılmaz (2014) define connectedness through the H-step generalized forecast-error variance decomposition (GFEVD) of an approximating VAR. The share of the forecast-error variance of variable i due to shocks in variable j is read as a weighted, directed link $j \to i$; row sums of the off-diagonal entries give how much each node receives ("from"), column sums how much it transmits ("to"), and the cross-sectional average is total connectedness. Using the generalized decomposition of Koop et al. (1996) and Pesaran and Shin (1998) makes the network invariant to the ordering of the variables. Rolling-window estimates track how connectedness changes over time.

NETS (Barigozzi and Brownlees, 2019) estimates the network directly as a sparse object: a lasso-type estimator of a VAR delivers a *Granger* network (lagged dependence), the sparse concentration matrix of the VAR innovations delivers a *contemporaneous* partial-correlation network, and their combination delivers a *long-run* partial-correlation network.

FNETS (Barigozzi et al., 2024) first removes a (possibly dynamic) common factor structure, estimated by dynamic principal components in a generalized dynamic factor model (Forni et al., 2000), and then estimates a sparse VAR and sparse (long-run) partial correlations for the idiosyncratic component. It therefore separates connectedness due to pervasive common shocks from idiosyncratic, network-type interdependence.

[To be completed: professor's papers on the idiosyncratic component, to be added]

1 Motivation

[To be completed: Why measuring connectedness among emerging equity markets matters (contagion, portfolio diversification, policy spillovers). Argentina is treated as one case among the 17 markets (its final role, decision 6 of paper structure v2, is still to be defined by the group). Suggested anchors from the data: the Argentine primary-election shock of August 2019 (largest weekly ARGT drop, -28.4% log return) and the March 2020 crash.]

2 Introduction

[To be completed: Research question (working version): how does measured connectedness among emerging equity markets change with the estimation method (Diebold–Yilmaz, sparse VAR, NETS, FNETS), and what share of it is a global common factor versus idiosyncratic? State hypotheses and the contribution; summarize the main results once Section 5 is written.]

3 Data

Universe and sample. We use US-listed, USD-denominated country exchange-traded funds (ETFs) as tradable proxies for 17 emerging equity markets: Argentina (ARGT), Chile (ECH), Brazil (EWZ), Mexico (EWW) and Peru (EPU) in Latin America; South Korea (EWY), Taiwan (EWT), India (EPI), China (FXI), Thailand (THD), Indonesia (EIDO), Malaysia (EWM), Vietnam (VNM) and the Philippines (EPHE) in Asia; South Africa (EZA), Turkey (TUR) and Poland (EPOL) in EMEA. The S&P 500 ETF (SPY) is included as a global reference node. Funds were selected if their price history starts by 3 March 2011 (7-day tolerance), at most 5% of weeks are missing, and average daily traded volume is at least USD 3 million. Colombian funds (ICOL, GXG), INDA and MCHI did not pass the criteria; India and China are represented by EPI and FXI. The common sample runs from 11 March 2011 to 23 October 2026, i.e. 816 weeks, with no missing values.